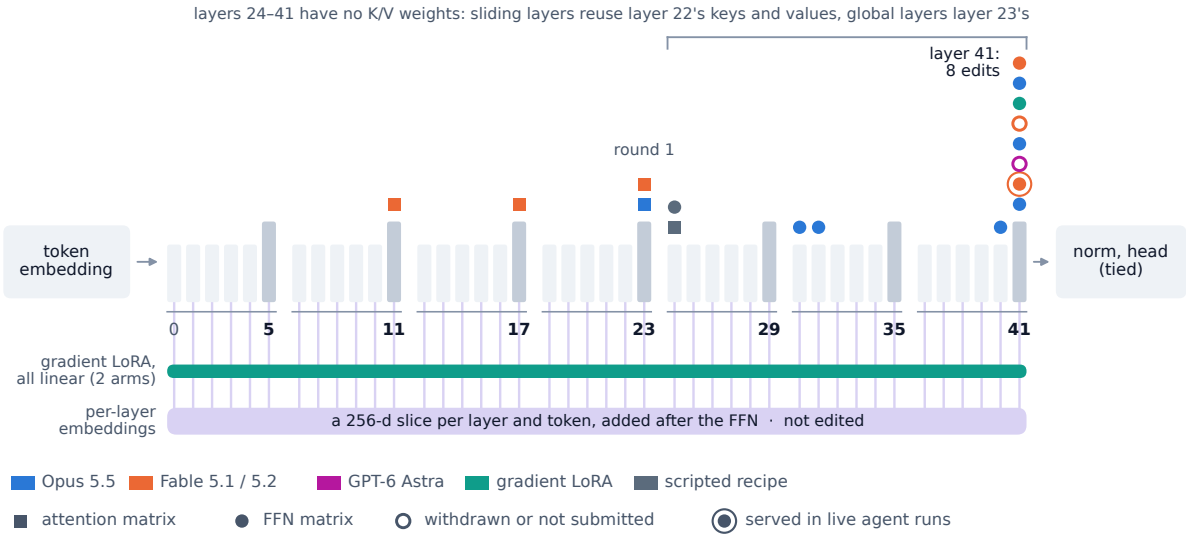
(a) Gemma 4 E4B: 42 decoder layers, seven groups of five sliding-window layers + one global



(b) Inside one decoder layer, in the order of the forward pass, with the edits on each matrix

Gradient LoRA (all linear, 2 arms): rank 8 on every q, o, gate, up and down in all 42 layers and on k, v in 0–23.

Order and shapes: transformers gemma4 forward pass and the E4B checkpoint (hidden 2560, FFN 10240). Edits: the frozen edit files.